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Archaeological Gameworld Affordances: A Grounded Theory of How Players Interpret Environmental Storytelling

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Abstract

Environmental storytelling is a design technique commonly used to convey narrative through assemblages of content in video games. To date there has been limited empirical work investigating how and on what basis players form interpretations about game environments. We report on a study in which participants (N=202) played a game about exploring a procedurally generated ruined village and were then surveyed on their interpretations. We draw on methods and theory from archaeology – a field that specialises in the interpretation of material remains – to support a grounded theory analysis of the survey responses, from which we form the theory of an *archaeological gameworld mental model*. Our study draws a novel link between affordance theory, archaeological knowledge production and game systems, and contributes new theoretical concepts that can be applied to procedurally generated and handcrafted methods in game design, narrative design and game preservation.

CCS Concepts

• **Applied computing** → **Arts and humanities**; • **Human-centered computing** → **User studies**.

Keywords

environmental storytelling, mental models, grounded theory, archaeological methods, affordance theory

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†References to this author may be made using the he/him masculine or they/them singular neutral pronouns.

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1 Introduction

In this paper we report on a study in which 202 people played a video game about exploring and interpreting a procedurally generated ruined village. We present the first extended grounded theory analysis of how players form theories based on environmental storytelling in a narrative-led game. Our resulting theory of an archaeological mental model of game worlds builds on the more general concept of a *mental model* (an internal framework for understanding external reality) which has been applied across a range of disciplines including cognitive psychology [45], HCI [14], and archaeology [25]. Intertwined with this is the study of *affordances*, which broadly refers to how an actor perceives the possible actions that can be taken with an object. The concept of affordances has been used to describe how game design can render potential actions perceptible to a player [54][11], and the potential of affordances to inform mental models in the wider field of design is also well established [67].

The study makes the following contributions to HCI research:

- We present the theory that players form *archaeological gameworld mental models*, based not just on in-game environmental storytelling but also the underlying game system. Pre-existing knowledge about the world and other games media also informs this model.
- Furthermore, players can be motivated to expand on this mental model through what we what call *archaeological gameworld affordances*, a novel way of framing how certain qualities such as ambiguity can encourage further player theorising.
- We argue that certain qualities of procedurally generated content, that we term *procedural affordances*, intersect with archaeological gameworld affordances, making them particularly evocative for environmental storytelling.
- We also discuss how *recording method affordances* can scaffold player archaeological gameworld mental models; recording aspects of their experience through methods such as map-making can help inform player interpretations and also be a method for eliciting player mental models in of themselves.

Overall, this theory builds on existing HCI models of interpretive agency and aggregation affordances, while also demonstrating the interdisciplinary potential of viewing player interpretations of environmental storytelling through the lens of archaeological theory. The archaeological gameworld mental model also has potential applications for designers and researchers interested in modelling player interpretations, prompting engagement through interpretive difficulty or combining grounded theory with iterative game design. We situate this work as a form of interpretive archaeology

[13], understanding that archaeological interpretation is a creative crafting of the past in the present, and that this can be a useful framing for how players form theories about fictional worlds in contemporary games.

2 Related Work

2.1 Definitions

In this subsection we briefly define terms and concepts that will be referred to throughout the rest of the paper.

Archaeology: Archaeology is the study of the human past through material remains. The discipline is traditionally associated with the excavation of ancient sites, however archaeological theory and methods have been applied to the contemporary past as well [39]. Archaeological methodologies are particularly well suited to interpreting human activity and intention through material remains, and in this study we are interested in how and on what basis players form interpretations based on environmental storytelling. As a design technique which expects players to intuit meaning through material remains, we set out to explore to what extent this process of interpretation has parallels with concepts from archaeological theory.

Archaeological Context: An archaeological context is, simply put “any single action, whether it leaves a positive or negative record” [69]; such as a deposit in the former case, or a cut into an earlier deposit, like a pit, in the latter. Archaeologists define contexts in terms of their relative position in space and time in order to develop a site chronology [68]. Thus, a pit cut into clay must have been dug later than the deposition of the clay, and any artefacts in the fill of that pit are likely to be associated with each other.

Archaeological Assemblage: The concept of “assemblage” is common across multiple fields, but it has a specific meaning in archaeology. It is usually used to describe a group of objects with a shared archaeological context [40], for example an assemblage of objects deposited together in a pit. The term is useful to refer to diverse objects with a shared chronological context, potentially revealing more about them as an aggregate, and the function of the context in which they were found.

Archaeogaming: Archaeogaming is a paradigm that emerged in the online archaeology community in the 2010s. It covers multiple intersections of archaeology and games; how heritage is represented in games, studying and reconstructing game code and the application of archaeological theory and method to games in recording them as archaeological sites [64]. Work in the field of archaeogaming includes studies of player-built structures in *No Man's Sky* [71], the reverse-engineering of the code and data for *Mystery House* [2], and the archaeological reconstruction of historical games using AI techniques [8].

Procedural Generation: Procedural content generation (PCG) is a term that refers to the creation of content, usually for video games, through the application of generative algorithms [76]. The term ‘generative’ here is used in the earlier, more literal sense of a process which generates something, rather than its contemporary appropriation to refer to large language models. The design, implementation and evaluation of generative systems across creative domains is a popular subject of research in computer science, but particularly within the practice of game development where it is

applied to achieve design goals [92], solve technical problems [22], and augment creative processes [81].

2.2 Environmental Storytelling

The coining of the term “environmental storytelling” is generally attributed to Don Carson in a 2000 article [15], applying his experience designing theme parks to video games. It refers to the staging of objects in a game world to suggest a story to a player.

2.2.1 Game Studies. Of particular relevance to our work is Don Carson’s use of “cause and effect” vignettes that “lead the game player to come to their own conclusions about a previous event.” Henry Jenkins [44] cites Carson in his taxonomy of environmental storytelling, the most relevant concept for our study being embedded narratives where the “game world becomes a kind of information space, a memory palace.” Fernández-Vara [30] refines the concept of environmental storytelling, introducing the concept of “indexical storytelling” that characterises the player not just as a visitor to a space but as an active agent or even detective who interprets and leaves traces in the game. This characterisation of the player as detective is taken up by Larsen and Schoenau-Fog [53] who class *Dark Souls* [32] as a non-detective game that nonetheless encourages players to interpret the past through environmental storytelling. Coles and Gillies [18] study of mixed-emotional experiences in games is also relevant to our study in that their grounded theory analysis identified *interpretive agency*, which is “the ability for the player to construct their own cognitive and emotional understanding” of the game. They identify environmental storytelling as contributing to interpretive agency, however they do not build on how and why this happens.

2.2.2 Archaeogaming. The link between archaeological interpretation and environmental storytelling in games is not new. Livingstone et al [55] introduced the concept of “archaeological storytelling” in their 2016 paper, in which they advocate for the use of material culture in a game world to tell a story. We appreciate the authors’ emphasis on going beyond singular stories or a preconceived “ground truth,” as in archaeology:

“A single ‘correct’ story may never be discoverable, instead it may be that a collection of plausible and possible stories may be found” [55].

Tara Copplestone’s [23] work on the development of archaeological games as a self-reflexive process allowing for non-linear storytelling and multiple interpretations echoes Livingstone et al’s sentiment above. In a 2022 GDC Smith Nicholls introduced “generative archaeology games” that encourage players to archaeologically interpret and record environmental storytelling [65]. This builds on Cook’s earlier proposal of “generative forensic games” as “a sub-genre of information games that challenge the player to understand the output of a generative system” [21]. *Nothing Beside Remains*, the game used in our study, could be characterised as a “generative archaeology game.” Caracciolo [9] has also discussed archaeological interpretation as a game mechanic that encourages indeterminacy through materiality. Bennett et al [6] have argued that players can take on an “archaeological mindset” when interpreting objects in *The Last of Us* [63].

More broadly the interdisciplinary methodological framework of Pearson and Shank's *Theatre/Archaeology* [70] is relevant, as it argues that both theatre and archaeology are site specific interpretive performances that require documentation to frame interpretation, and the same can be applied to games. To date, archaeogaming work has not produced an extensive empirical study of how players themselves potentially apply archaeological theory and method in order to interpret environmental storytelling in games.

2.2.3 Procedural Content Generation. Though there has been considerable work on procedural level generation and procedural narrative, there has been limited work that connects these sub-domains, especially in terms of qualitative work on players' response to procedural environmental storytelling. Nielsen et al conducted a quantitative study of player responses to procedurally generated environments [66] which has parallels with our study but is more focused on successfully conveying key narrative information than the process of how players form those interpretations. Sandhu and McCoy [74] have recently published a survey of the field in which they propose creating a procedural indexing system, taking inspiration from Fernández-Vara's aforementioned work. They also cite our work presenting the preliminary results of the present study [79], which argues for an archaeological approach to the emergent storytelling possibilities of procedurally generated content.

Work in the sub-domain of procedural narrative, while not necessarily engaging with environmental storytelling, is especially relevant in terms of how players construct emergent narratives. James Ryan's [72] curationist approach to PCG has been particularly influential. There are a range of different proposed qualitative and quantitative approaches to evaluating procedural narrative, and the most relevant for our survey is Eladhari's [28] work on retellings. She considers player retellings to be an indicator of a well-designed interactive narrative system on the basis that it indicates players encountered memorable and meaningful stories. Kreminski et al [51] have drawn on this in their grounded theory analysis of AI-based game retellings, in which they identify players engage in "extrapolative narrativization" to fashion "additional story elements that are not directly modeled within the game's systems."

2.3 Mental Models

In *The Psychology of Everyday Things*, Norman [67] defines mental models as "the conceptual models in people's minds that represent their understanding of how things work." The concept of mental models has a long history in cognitive psychology [45], especially in terms of spatial reasoning [85], as well as HCI. [14]

2.3.1 Mental Models in Archaeology. Rather than trying to intuit the thought processes of players, archaeologists try to understand the cognition of past peoples through mental models. The sub-discipline of cognitive archaeology, broadly speaking, attempts to infer past cognition through material remains [25]. However, in terms of our own player study we are more interested in mental models as they relate to archaeological knowledge production. Barceló's work [5] is somewhat relevant here in that it focuses on the requirements for an "automatic archaeologist" AI agent to be able to infer cause and effect from material remains. Morgan et

al's work on how archaeologists' mental models are buttressed by different drawing methods is highly applicable to our work. As they put it:

The affordances of the tool and how it interacts with our bodies, the environment, and the interpretive subject combine to create a mental model of the archaeological remains. [62]

Though not strictly speaking concerned with mental models, we also consider Shanks' work on *The Archaeological Imagination* [77] to be relevant as it is concerned with creative responses to the past, and how archaeological knowledge production is also shaped by wider popular conceptions of the discipline.

2.3.2 Mental Models in Game Studies. Mental models have been applied widely in game studies, usually as a way of trying to represent changing player knowledge and perception. This is the case with Williams' study on players' use of different video game controllers [90]. Banks and Bowman's work [4] understands video game avatars to not be just singular bodies, but assemblages conceived of through player mental models. Of note is Ubochi's work [86] on using mental model elicitation techniques to better understand how players conceive of complex game systems. Furthermore, Cardona-Rivera [12] conceives of mental-model building as a desired interpretation outcome of narrative design, getting the player to "mentally simulate a real or possible world."

2.3.3 Mental Models in PCG. Though with a broader focus than PCG, there have been a number of recent qualitative studies on how players form mental models of AI agents in games. Gero et al [35] have published work on mental models of agents in a cooperative game, while conversely Villareale [88] have looked at mental models of adversarial agents, identifying how these models shift over time and after the acquisition of new information. Cimolino et al [17] conducted a grounded theory study on non-gamers' experience of automation confusion, identifying player behaviour as the result of inaccurate mental models. Though it does not explicitly reference mental models, Watson's work on "procedural elaboration" [89] in *Minecraft* [60] is also applicable as it refers to players attempting to understand the underlying generative system of the game.

2.4 Affordance Theory

The term "affordance" was originally coined by Gibson [36] in the domain of ecological psychology, with affordances of the environment being: "what it offers the animal, what it provides or furnishes." In 1988, Norman [67] published *The Psychology of Everyday Things*, applying it to the design of everyday objects, which subsequently influenced early work on "technology affordances" of user interfaces [34]. Crucially, Norman disagrees with Gibson's idea of affordances existing irrespective of whether they are perceived or not, instead emphasising the importance of designing for what users perceive as possible instead.

2.4.1 Affordances in Archaeology. Affordance theory has not been extensively engaged with within the archaeogaming literature, with some exceptions. Bozdog [7] has done research into site-specific performances that are informed by the affordances of a heritage site, while Copplestone [24] has examined the imposition of "external

affordances” on how narratives should manifest in a media form have limited their application in heritage settings.

As a discipline concerned with the material culture of the past, archaeology contends with object ontologies, including their multiple temporal and spatial affordances as they relate to assemblages [49]. Tim Ingold [41] criticises the tension in Gibson’s original theory that objects have inherent properties, instead arguing that affordances only exist as they are perceived. At the same time, he is skeptical of a cognitivist approach that cannot conceive of meaning unmediated by semiotics. He advocates for thinking about affordances “in common,” that are constructed in public and through interaction between people and objects. Knappett [48] takes a relational stance on ecological psychology and situated cognition, seeing them as interdependent.

Archaeologists have used digital technologies such as GIS and 3D modelling to attempt to understand experiential and affective affordances of archaeological sites [37][29]. There has been much recent discussion of digital materiality in archaeology [61] and especially the affordances of digital objects in museums [1].

2.4.2 Affordances in Game Studies. There has been extensive work on affordances in game studies and game design, drawing from different theoretical traditions. Linderoth [54] follows Gibson’s ecological model, arguing that games are not well-suited as learning environments but instead can be effective at conveying exploratory and performatory affordances.

Conversely, Mateas [57] follows Norman with an emphasis on providing a sense of player agency through balancing affordances. Cardona-Rivera et al [11] propose a cognitivist theory of affordances in games which that they conceive present real and perceived affordances. Of particular relevance to our study is Young and Cardona-Rivera’s work [91] on narrative affordances in games, in which they define the term as the ability for a player to envisage how actions will contribute to the future completion of a storyline. Also especially relevant is Dugas et al’s [84] survey of map interfaces in games, in which they identify “atomic affordances” as individual interface components that can be manipulated by players. They also found that the ability for players to make their own records through drawing and text support on maps was particularly useful for puzzle-solving. More broadly in HCI studies, Kaptelinin and Nardi [47] have critiqued Gibson’s model as not being able to account for “technologies as a special type of object,” with the need for more precise terminology such as “aggregation affordances” that refer to how technological artefacts can be combined with other (digital and non-digital) artefacts. In her work on gameworld affordances, Jørgensen[46] also considers that Gibson’s model is inadequate for game designers who need to be cognisant of perceived rather than just innate affordances. Furthermore, we find Jørgensen’s definition of a gameworld as being both the fictional designed space for play, and the underlying system that operates it, to be useful for our subsequent analysis.

2.4.3 Affordances in PCG. Affordances have been discussed in relation to procedural content generation in the academic literature, though often as a form of taxonomisation. For example, Summerville et al [82] refer to game affordances in terms of having a PCG system understand the semantic properties of game objects, while Sarkar et al [75] develop an affordance vocabulary to describe

the properties of different generated tiles. Smith [78] refers to the affordances of different deployments of PCG in games from a design perspective. Though it relates to artificial intelligence research and art more generally, Mateas’ [58] work on interpretive affordances are particularly relevant to the present study in that they “support the interpretations an audience makes about the operations of an AI system.” Though it doesn’t explicitly mention affordances, Karth’s [58] *Preliminary Poetics of Procedural Generation in Games* acknowledges that in existing taxonomies of PCG “the focus has been on the how rather than the why,” instead attempting to chart generative aesthetics and how they complement each other. Kreminski [50] has also reflected on the hauntological experience of playing games with PCG that are haunted by their own past cycles of content. Furthermore, Cardona-Rivera’s [10] thesis on interactive narrative affordances specifically focuses on how an AI system can model a player’s understanding of those affordances.

3 Nothing Beside Remains

Nothing Beside Remains (NBR) is an independent video game originally designed by the second author in 2018. It is a 2D, single player exploration game set in a ruined village that uses ASCII symbols to represent objects and environmental features, inspired by terminal-based roguelikes from the 1980s and 1990s. Figure 1 shows two screenshots from the original version of the game. The player, represented by an “@” symbol, can move around using the keyboard arrow keys. If the player walks into an object, a text box will appear with descriptive text. The game has no win condition or end point, and no explicit goal.

Each village is procedurally generated, with two phases of simulation; one which abstracts the life of the village pre-ruin and its progress towards one of three endings: famine, being overrun by wild animals, or ecosystem collapse. Each of these endings then feeds into a second phase of simulation that effects details of the final generated village that a player actually explores. The generator was also designed to give the impression of more nebulous cultural practises, such as a random number denoting the number of legs a chair has. Despite variations between different generated villages, they always have several features in common: a large religious building, four lakes at the corner of the map, and a collapsed statue where the player begins.

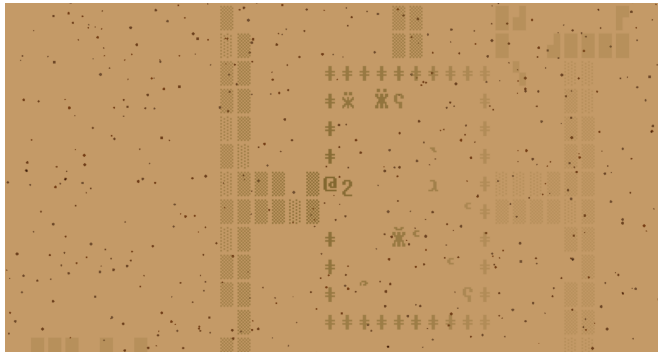
In 2022, the first author extended NBR with new content, augmenting its generator with new features, written descriptions and patterns, as well as changing the meaning of some of its generated text. This version of the game forms the basis for both the initial study and theoretical sampling we conducted, both of which we describe in the next section.

4 Method

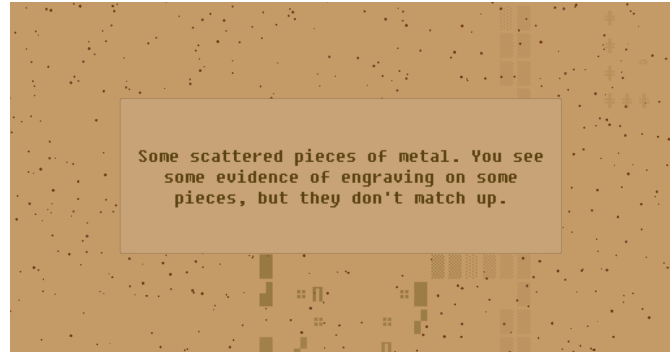
In this section we describe our methodology for two online surveys, the results of which were subsequently subject to a grounded theory analysis. The initial survey was conducted in 2023, and the subsequent theoretical sampling was conducted in 2024.

4.1 Grounded Theory

We chose to conduct a grounded theory analysis of player responses to procedurally generated content for several reasons.



(a) The player (represented by an @) is standing at the end of a path, in the entrance to a field.



(b) When the player walks into an object, a textual description appears.

Figure 1: Two screenshots from the original version of *Nothing Beside Remains*

Firstly, grounded theory is particularly well-suited to work in nascent fields where limited or no theories exist [87]. While grounded theory has been applied extensively to the study of games and HCI more generally, it has had limited application in the analysis of procedural content generation [51] and the archaeological study of video games [26]. The line-by-line coding of grounded theory allowed us to engage with the rich written interpretations of respondents. Furthermore, and as will be further detailed in the section on Theoretical Sampling, the iterative nature of grounded theory analysis complemented our own iterative design of *Nothing Beside Remains*.

Cognisant of Salisbury and Cole’s lament that much work refers to grounded theory in a generic way [73], we follow a constructivist grounded theory approach [16] based on abductive reasoning [20]. The chief reason for choosing this strand of grounded theory is the constructivist emphasis on a reflexive approach that acknowledges a researcher’s pre-existing experience and beliefs inevitably influence their reading of the data. Unlike classical grounded theory [38] which encourages the researcher not to engage with relevant literature prior to analysis, we did this to be guided by “sensitising concepts” [16] to frame the study. With this in mind, we began with the following open-ended research questions:

- RQ1: How do players form interpretations based on environmental storytelling in *Nothing Beside Remains*?
- RQ2: How does the generative design of *Nothing Beside Remains* influence player interpretations?

We followed Charmaz [16] in aiming to produce several core theoretical concepts woven into an interpretive narrative that takes into account implicit meanings and social context.

4.1.1 Coding Process. We followed the iterative process of initial coding, focused coding and theoretical coding as laid out by Charmaz and Urquhart [16][87]. This process is illustrated in Figure 2. Due to the large volume of respondents for the initial survey, these were coded after the survey closed. In the initial coding phase, we chose to code the answers in themed clusters, so questions 0, 3 and 5 were coded together as they pertained to procedural content generation, questions 1, 2 and 4 were coded together as they pertained to interpretation, and finally question 6b was coded separately as

that pertained to potential recording methods. There was constant comparison both within and between these clusters.

In the initial coding phase, we conducted line by line coding. In the focused coding stage, constant comparison between codes led to the formation of core categories. In the theoretical coding stage, we wanted to understand how these core categories related to each other, and used the “clustering” [87] technique that Charmaz describes, which is essentially a form of rapid mind-mapping to visualise relationships between codes or categories. From this process, we had the initial working theory of an archaeological mental model, that we went on to test in the theoretical sampling. Memos were made in all three coding cycles to reflect on the process and make note of emerging ideas, concepts and relationships.

The first author was the sole coder. Charmaz is critical of Strauss’ [80] push towards verification with grounded theory, instead advocating that “Rather than contributing verified knowledge, I see grounded theorists as offering plausible accounts” [87]. For this reason, and in the spirit of the constructivist tradition, we leaned into the subjective coding of a single researcher.

4.2 Positionality Statement

4.2.1 First Author. I am a PhD researcher in a computer science department, however I worked as an archaeologist for roughly 7 years before I started my studies, and much of my research is informed by this background. Furthermore, since 2016 I have been doing work in the emerging field of archaeogaming. I was motivated to study player interpretations of environmental storytelling in games as I observed that the implicit narrative techniques it uses have parallels with archaeological theory-making. As the sole coder for the grounded theory study, I am aware that I brought my own experience as an archaeologist to the analysis and applied technical archaeological terms. Though of course this reflects my inherent bias, from a constructivist point of view it is useful to be able to trace the influence of my academic and professional background on how the codes were formed.

Furthermore, I have worked in the writer’s room of an indie game studio, thus I have some insight into the kinds of narrative design conversations that happen in commercial games studios. As a queer non-binary person, I’m more broadly interested in queer

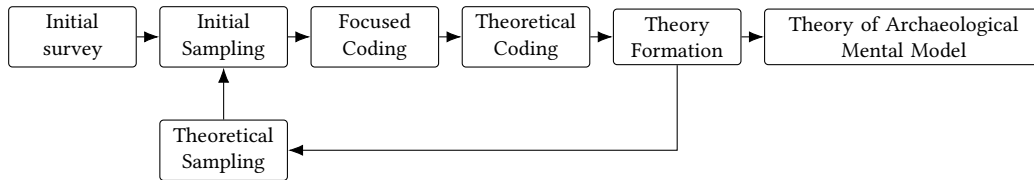


Figure 2: The coding process adopted during our work.

game studies and the queer potential of uncanny procedurally generated content, which influenced my perspective on *Nothing Beside Remains*.

4.2.2 Second Author. I am an artificial intelligence researcher by training, with a focus on quantitative analysis techniques and the study of generative systems. As such, I am most used to interpreting player experiences through recordings of player input data and performance metrics. This certainly shaped my initial involvement in discussions with the first author about this experimental setup. The first author suggested using grounded theory as an analytical technique, and was fully responsible for the coding of responses, and so the capacity for my background to bias the coding process was limited. I have supervised and helped run qualitative user studies using a variety of methods, including structured and unstructured interviews, and this helped inform my contribution to the work outlined here.

I have played, made, researched and written about games for the entirety of my career, including those made in independent and art scenes, similar to the first author. This helped me maintain a broad perspective both on our work with *Nothing Beside Remains*, and the implications of our work on games beyond the normative ideal of a commercial video game-as-product. I am also an experienced researcher in the field of generative systems, having contributed towards the theory and practice of both designing and understanding generative systems for games, which helped contribute additional perspective and understanding on what makes generative design a unique topic for a study such as this.

4.3 Procedure

4.3.1 Nothing Beside Remains as Case Study. We chose *Nothing Beside Remains* as the sole game for our study for several reasons. Firstly, as the developers of the game it allowed us to create bespoke builds for the survey. This was especially important as the game has procedurally generated content, so having the ability to dictate which seeds players would access gave us a baseline for understanding what material culture players could potentially encounter. This also allowed us to collect telemetric data of what routes players took, and what objects they interacted with, which we will analyse in a subsequent complimentary quantitative study. Secondly, as the developers we have insight into the design goals and development of the game that we would not otherwise have. This means we can bring our own knowledge of the generator that underlies the gameworld to the subsequent analysis. *Nothing Beside Remains* was essentially a small petri dish for our experiment, heavily constrained but allowing us to analyse player interpretations

at a much granular level than if they were exploring a larger commercial game. Thirdly, as a very short abstract game hosted on the web, *Nothing Beside Remains* was easily accessible to participants in an online survey, with the only hardware requirement being access to a computer with an internet connection. It is also requires a limited time investment to play. That being said, we acknowledge the limitations of our sample, which is indicative of responses to an abstract 2D game designed for emergent storytelling without a comparative case study.

We created a bespoke build of *Nothing Beside Remains* for the study that gave us more control over the configuration experienced by players. In the first seed, depicted in Figure 3, the village is partitioned into two parts, separated by a large path, with the religious building placed more centrally. In the second seed, shown in Figure 4, the village is densely packed with the player starting directly beneath the only flower field. Water features are plentiful and large, and the religious building is towards the edge of the village map. We felt both villages had distinct qualities that would allow us to see how participants responded to two distinct outputs of the generator.

After generating the village with a 50% chance of picking either seed, our version of *Nothing Beside Remains* then has a 50% chance to place an additional item in the village - a pocket watch. This hand-written item was deliberately included in order to potentially prompt players to question why a mechanical object anachronistic with the other material culture represented in the world was present. If the pocket watch is added, it is generated next to the altar in the religious building. Its description is as follows: "A pocket watch made of a delicate treacle coloured metal is embedded in the dirt. There are deep scratches and a stain partly obscuring the face. You can make out part of an inscription: DON." Its in-game visual representation is a unicode symbol of a lowercase delta: δ .

4.3.2 Survey Design. Generally, grounded theory analyses in game studies have utilised semi-structured small interview samples [3][19][17], however we used an online survey with set questions. This form of data collection falls under what Charmaz characterises as "elicited texts" [16]. As Charmaz herself identifies, this form of data collection has the advantage of potentially allowing participants to be more honest in their answers than if they were involved in an in-person interview. This point is particularly salient given that participants were aware the game they were playing was one the authors themselves had made, so there may have been a perceived pressure to be complimentary. Further to this point, we found that participants were comfortable being blunt about their dissatisfaction or disinterest in the game. To give an example, when asked about potentially recording the game one participant responded that they would not:

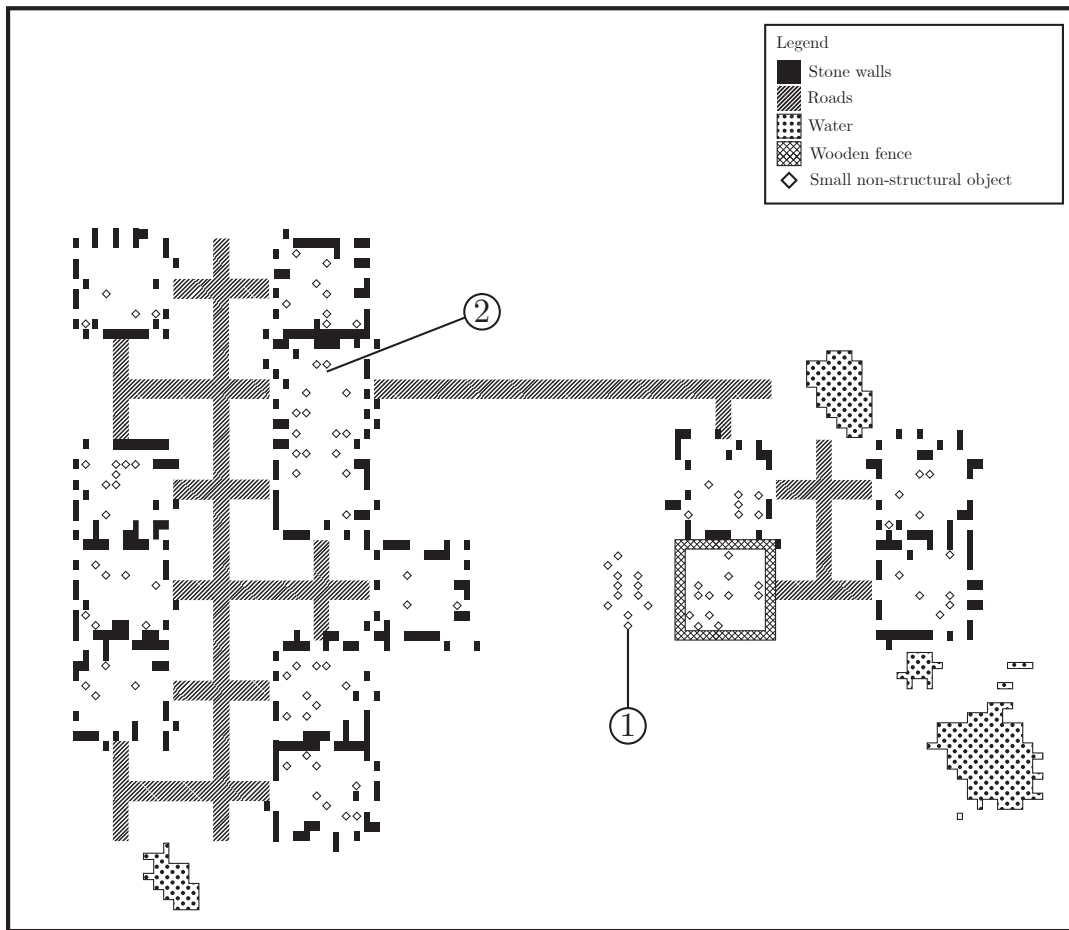


Figure 3: One of two village seeds used in the initial study. The player spawns at the point marked 1, next to the scattered remains of the statue. The pocket watch, if present in the village, is at the point marked 2 next to the altar in the religious building.

“Because I don’t think it would be worthy of recording or interesting for others to read/see/hear.” (P127, Q6b)

None of the survey questions, aside from those asking if the participants had played the game before, were mandatory. This meant that participants could opt-in to giving interpretive answers with as much or little detail as they wanted. The survey itself was split into three parts. The first part asked for participant consent. In the second part, participants could access a bespoke build of *Nothing Beside Remains* in a separate window. This build was made so that we had control over what seeds of the game players could access and collect their play data. Cross-referencing between survey responses and play data was made possible through participants entering a personalised in-game code into the online form. The third section had one mandatory question asking if participants had played the game before, as the original has been online since 2018 and player interpretations could be affected by previous playthroughs. We also asked several demographic questions regarding participants’ interest in games as a hobby, if they worked in the games industry or used games as part of their academic research. Furthermore,

participants were asked to define the term “procedural content generation” so we had a sense of how they themselves understood it in their own words. The remainder of the third section contained six non-mandatory questions with free text responses. The free-text questions relevant to our coding and grounded theory are shown in Table 1, while a full list of survey questions is in Appendix A.

4.4 Participants

In the initial survey, we received responses from 187 participants over five days. Of those participants who volunteered to put their age in the survey form, the average was 35 (st.dev. 9.6 years). Out of these respondents, 174 declared they play games regularly (92%), 36 identified as working in the games industry (19%) and 35 as working in games research (18%), with some participants declared as both. As stated in section 4.3 above, participants were randomly assigned one of two seeds either with or without the hand-placed pocketwatch object, leading to four possible variations. Overall, 87 respondents played the first seed, 97 played the second seed, and three respondents could not have their seed determined after the

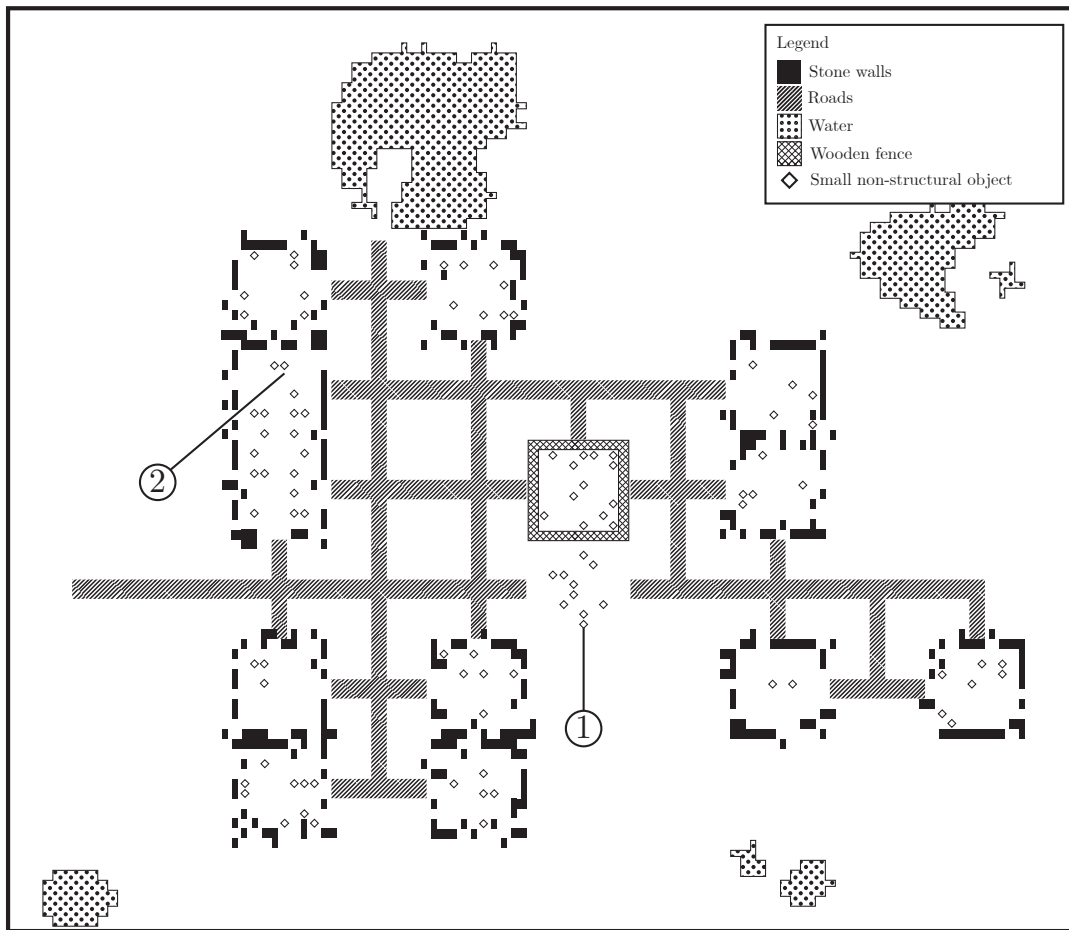


Figure 4: One of two village seeds used in the initial study. The player spawns at the point marked 1, next to the scattered remains of the statue. The pocket watch, if present in the village, is at the point marked 2 next to the altar in the religious building.

Code	Question Text
Q0	How would you define “procedural content generation”? If you are not familiar with the term you can leave this field blank.
Q1	Write your interpretation of what happened to the village in the game. What reasons do you have for this interpretation?
Q2	Do you have any alternative interpretations? If so, explain your reasoning.
Q3	Could you distinguish between procedurally generated and hand-written content, and if so how?
Q4	Pick one object from the game and describe how you think it came to be left where you found it in the village. Why did you choose this object?
Q5	Do you think the object you chose for question 4 was procedurally generated? Why or why not?
Q6a	If you were to create a record of your experience in the game, which of the following methods would most appeal to you: Creative writing; Screenshots; Video footage; Map-making; Other (specify)
Q6b	Why did you choose the above method(s)? If you were to share the record with others would that change your answer?
Q6-2	If you made a record of your experience playing the game, what method did you use and why? Did it affect your experience or interpretation of the game?

Table 1: All survey questions. In the theoretical sampling, questions 6a and 6b were replaced with the question labelled here as 6-2.

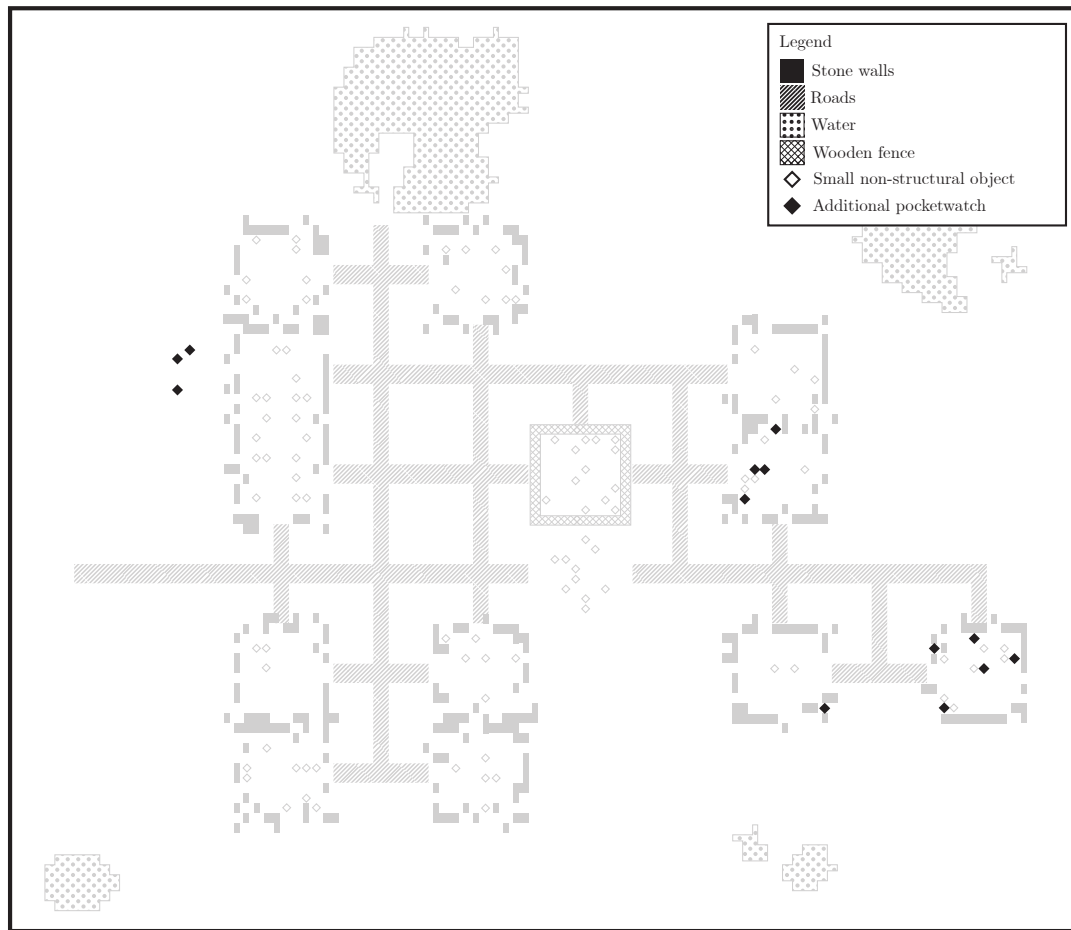


Figure 5: The positions of the additional pocketwatches added for the theoretical sampling. This is a modified version of the map shown in Figure 4, with the original map lightened to better show the location of the watches.

fact due to an initial server error which was quickly fixed. Furthermore, 102 players played the condition with the pocketwatch and 85 did not not.

4.5 Recruitment

Participants were recruited through the two authors promoting it through their own personal social media accounts on Twitter, Mastodon, Discord, TikTok and Facebook. This method of recruitment does hold some obvious drawbacks in that it is limited to the authors' online social circle of peers who already have domain knowledge of archaeology and game development, however these are also arguably groups that are the most likely to be interested in and benefit from the research outputs of the study. We collected demographic information related to participants' involvement with the games industry and heritage sector in order to get a better idea of how this was represented in the sample. The only exclusion criteria for the study was that participants had to be over the age of sixteen. For the initial survey, we aimed to get at least one hundred respondents for a statistically meaningful sample size, and collected data between the 20th and 25th of January 2023.

4.6 Ethics

Both the initial study¹ and the theoretical sampling² were subject to the King's College London minimal risk self-registration process. The studies were conducted anonymously through a Google form that did not collect any identifying or sensitive information. Participants were provided with an Information Sheet³ that included extra context on the research aims of the project, its data handling policy and stated that data about their movement and interactions in the game would be collected as part of the study. Participants could not take part and access the bespoke build of the game for the study without providing their informed consent in the first part of the Google form.

4.7 Theoretical Saturation

4.7.1 Participants. The follow-up survey for theoretical sampling was open to submissions between the 27th and 30th August 2024. For this survey, we were not aiming for a specific sample size but for

¹Ethical review reference number: MRA-22/23-35186

²Ethical review reference number: MRA-23/24-45362

³possibilityspace.org/nbrstudy/information-sheet-nbr-study.pdf

theoretical saturation. We analysed responses from 15 participants. Of the participants who volunteered their age, the average was 31.7 (st.dev. 10.5 years). Out of these respondents, 14 respondents stated that they play games regularly (93%), 4 identified as working in the games industry (27%) and 2 as working in games research (13%). In addition to this, 6 stated they had a personal interest in heritage or archaeology (40%) with some of these participants also identifying as working in the games industry. 3 participants said they often recorded their game play experiences in some form (20%). Of the fifteen respondents, seven received the condition with a single pocket watch (47%), seven received the condition with multiple pocket watches (47%), and one could not be determined due to inaccuracy in the survey entry.

4.7.2 Survey Design. For the follow-up theoretical sample, we made the following changes to the survey:

- We added the option for participants to declare they had an interest in archaeology and heritage, worked in the heritage industry, and studied the past in some form in the demographic section. We had not accounted for this in the initial survey.
- Participants could also declare they regularly recorded their gameplay experiences as this would potentially add extra context to question 6.
- Participants were explicitly encouraged to record their gameplay experience via this instruction in the form: “We encourage you to make a record of what you find in the game – how you do this is up to you.”
- Question 6 was changed to “If you made a record of your experience playing the game, what method did you use and why? Did it affect your experience or interpretation of the game?”

The aim of making these changes to the survey was to better understand the pre-existing knowledge of our survey participants, and also to see if our working theory, especially that of recording method affordances, could accommodate respondents who were actually primed to record their game experience.

4.7.3 Iterating on *Nothing Beside Remains*. We also prepared a new version of *Nothing Beside Remains* for use in the theoretical sampling phase. Unlike the first phase, this version of the game uses a single seed – the layout of which can be seen in Figure 3. The game will generate one of two versions of this seed with equal probability. The first is identical to the pocket watch condition in the original study, with a single pocket watch fixed to appear next to the altar in the village’s religious building. In the second, the pocket watch is placed next to the altar as before, however in addition to this there is a 60% chance that a pocket watch will also be generated next to any skeleton object that is spawned in the village. Unlike the original pocket watch, these additional items have generated descriptions using a grammar-based template similar to the rest of the game’s generated text. Because the generation and placement of these pocket watches is linked directly to the seeded random number generator used to create the village, all players receiving this second condition with multiple pocket watches saw the same output.

We made these changes to *Nothing Beside Remains* in line with our working theory of gameworld affordances, specifically assemblage affordances, as the new pocket watches were designed to generate near skeletons, which could be conceived as associated assemblages. Furthermore, the multiple near-identical pocket watches were deliberately designed to be incongruous and ambiguous, thus also drawing on the archaeological gameworld affordances we had already identified at this point in the analysis.

5 Results

Initial coding resulted in 335 unique codes for the first survey, and 119 unique codes for the theoretical sampling. Codes were recorded in a spreadsheet along with memos to document the first author’s process. Through several iterations in the selective coding phase, initial codes were grouped under broader concepts, as can be seen in the table in Appendix B. In this section we summarise the theory that the first author arrived at following theoretical sampling and the conclusion to the final theory formation. The diagram of this theory, showing the concepts and how they relate to one another, can be seen in Figure 6. The theory centers around our proposal of an *archaeological gameworld mental model*. The remainder of this paper contains direct quotes of participant responses; they are followed by a note indicating their participant number (e.g. P20) and the question they are answering in the quoted passage (e.g. Q4). Note that the question to define procedural content generation appeared before the main part of the survey, and has hence been numbered Q0. We highlight sections of quotes in bold for emphasis.

5.1 Archaeological Gameworld Mental Model

The core category, player archaeological gameworld mental models of games, is characterised by four sub-categories that will be elaborated on below:

- *Pre-existing knowledge* that *informs* the mental model;
- *Gameworld affordances* that *constrain* the mental model;
- *Archaeological gameworld affordances* that *inspire elaboration* of the mental model;
- *Recording method affordances* that *scaffold* the mental model

We use the term ‘gameworld’ in our model as Jørgensen [46] defines it to refer to both the playable game space and the underlying system that operates it. The catalyst for this core category was participants’ explicit referral to their own internal conceptualisation of the game and their theories about it:

“Screenshots and video recordings don’t do it quite justice since **part of the experience takes place in the mind**, not just on the screen.” (P26, Q6b)

“**Almost all of the experience was mental**, taking in the objects I saw and forming my own picture of the village and what could have happened.” (P167, Q6b)

“I think the abstract style of the game lends it to some sort of **theatre of the mind** where the player could write and describe their interpretation with screenshots and/or a map for reference.” (P170, Q6b)

Participants’ expression of their own personal mental models reflects the literature on player mental models that is unique to each

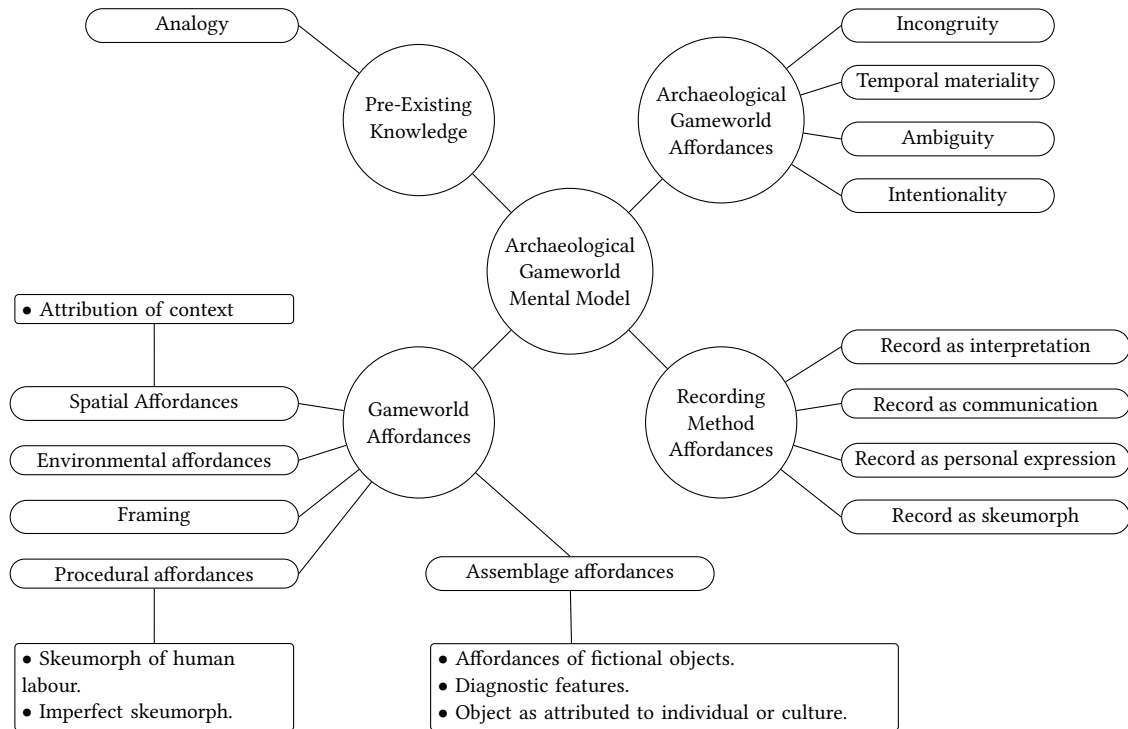


Figure 6: Grounded theory of archaeological gameworld mental models in games

individual [88]. One of the ways in which player mental models are particularly hard to predict is that each individual approaches a game with different pre-existing knowledge.

5.2 Pre-Existing Knowledge

Pre-existing knowledge is a category that broadly refers to how players bring their own understanding of the world and knowledge of games to their play experience. One of the clearest ways in which players expressed how pre-existing knowledge informed their interpretations of *Nothing Beside Remains* was through referring to or making analogies with other games. For example, when asked to define procedural generation, one participant responded:

“Something that is not pre-determined by the designer, but the mechanism for its creation is determined by the designer. In my opinion, stitching together from pre-defined building blocks also qualifies (see *Spelunky 2 level generation*)” (P80, Q0)

When asked if they could distinguish between hand-written and procedurally generated content, another participant referred to their knowledge of how terrain generation works in *Minecraft* [60]:

“It would depend on the setting. Procedural content tends to show patterns, as it would need to be programmed according to a set of rules. **So, for example, a grassland biome in a Minecraft world will always have a relatively flat landscape, and a lot of oak trees.**” (P97, Q3)

Just as players would refer to iconic PCG-driven games when discussing PCG, they would also refer to iconic archaeological sites when asked about their interpretation of what happened to the village. Several participants made analogies with Pompeii:

“The low howling wind tells me whatever happened is still going on, though maybe less intensely. The presence of surface skeletons makes me think it was a rapid event. **Like Pompeii, but somehow with sand instead of ash.** I can’t explain it.” (P3, Q1)

“Many of the houses do appear to be totally empty... This implies that many of the residents had already abandoned their village, with the remainder choosing to stay inside. Perhaps they knew something was coming? **Either because it was slow, as in my initial starvation hypothesis, or because they evacuated, as was the case for many of the luckier residents of Pompeii.**” (P186, Q2)

Players also expressed how a *lack* of pre-existing domain knowledge likely influenced their interpretations:

“I don’t think I could; **I’m not familiar enough with the difference or capabilities of procedurally generated content.** I think the statue description was hand-written because it was more specific.” (P145, Q3)

5.3 Gameworld Affordances

The subcategory of gameworld affordances contains many further nested subcategories that can constrain the archaeological gameworld mental model of the player, guiding them towards a potential interpretation over others. In some cases the first author drew from archaeological terminology when coding these categories, such as “attribution of context.” In the below example a participant implicitly infers the reason an object was left in a particular location from its assumed context:

“The sickle which was left in the fenced off area with plants (probably a farm of some kind). It was left there because it was probably used for farming, and I chose it because this was the area I saw (before getting stuck at the bottom of the map) that most made sense to me.” (P135, Q4)

Assemblage is another term with a more specific meaning in archaeology; it refers to a group of artefacts in the same context, associated with each other. There are numerous examples of participants ascribing meaning through the association and proximity of objects:

“I choose it because finding all the skeletons in one place felt like an interesting clue to what happened to the village, but it wasn’t initially clear whether there were humans, or animals, or what, and finding a large clearly animal skeleton recontextualized the collection of skeletons and made me consider that maybe this was something like a barn and all the skeletons were from animals that were penned up here.” (P154, Q4)

“The seven-legged chair was in a room with a few other chairs, so I suspect the chamber was a living area of some sort. I chose it because 7 is a memorable number as far as chair legs are concerned.” (P68, Q4)

Furthermore, participants’ mental models were influenced by the assumed affordances of fictional objects:

“something happened here which caused the residents to leave. They appeared to have plenty of time to leave but also had to do so without taking anything too heavy. This implies to me they may have been suffering a severe drought or some other break down of their local ecology, thus leaving for better places to live.” (P184, Q1)

“The fact that many of the objects left behind were ones that would not be easily carried (heavy ceramics) or would not be practical on a journey suggests that the people left rather than dying out.” (P86, Q1)

Participants assumed that certain objects would have been too heavy to carry and incorporated this assumption into their own internal mental model of what they think happened to the fictional village and how they believe its inhabitants would have behaved as a result, leading to the resulting material remains found in the game landscape.

5.3.1 Procedural Affordances. Although the subcategory of gameworld affordances can be applied generally, we also identified a subcategory within this that relates specifically to the procedurally generated nature of the game under study, which we term *procedural affordances*. The code of “imperfect skeuomorph” refers to participants believing that PCG is characterised as an attempt to imitate human labour which can be identified through its strange syntax and other uncanny features:

“If a large enough amount of the procedurally generated content can be observed specific patterns can be found, such as similar phrasing and ‘mad libs’-like word substitution patterns. Hand-written content on the other hand is usually very distinct from each other and whatever similarities exist are more thematic or nuanced.” (P99, Q3)

“Using ‘the computers’ to create a very large, varied set of objects (or maps, images, sounds etc) that would be difficult or impossible to create manually. But shepherded by a human who gives direction and adds constraints.” (P105, Q0)

“Algorithmically generating content (often used in a game context but is useful for far more applications) that, ideally, is of good enough quality that it appears that humans created the content, as opposed to random-stuff-in-random-places.” (P30, Q0)

“PCG is a system to extend human intent beyond human effort.” (P64, Q0)

These procedural affordances are particularly powerful as they compliment another subcategory of the archaeological gameworld mental model – archaeological gameworld affordances.

5.4 Archaeological Gameworld Affordances

Archaeological gameworld affordances are qualities that invite further elaboration on the archaeological gameworld mental model by providing further evidence for the model or challenging existing assumptions. Here, we discuss two of the four identified affordances. The first of these is *incongruity*. When participants encountered material culture that didn’t fit their existing mental model, it motivated them to speculate further:

“I think that the climate must have changed and the village became uninhabitable. There are altars and jugs for water, but no bodies of water in this village. The landscape seems harsh and not as welcoming for human life.” (P145, Q1)

“The bucket in the muck. I chose this because mostly because it was confusing (what is muck in the context of desert?)” (P59, Q1)

Another key archaeological gameworld affordance is *intentionality*, which manifested in various ways. In some cases, participants’ identified apparent intentionality in the material culture of the fictional village:

“In one of the houses there’s a stone altar that had a metal bowl (or jug) that was split into two. I think the splitting was done deliberately, for a religious

reason. It would be highly unlikely for such splitting of metal to happen accidentally." (P137, Q4)

"The pocket watch I found in the large ceremonial structure. **it seems to be a personal object, it was possibly dropped or it might have been left intentionally as a religious object.**" (P82, Q4)

"Ok yeah, the Pocketwatch Apocalypse. **Everyone got addicted to these things.** They don't tell time, but they tell you something, and some have creepy vibes." (P193, Q2)

Intentionality also relates to how participants ascribed this concept to content that they felt was perceptibly unique and hand-written, thus they did not ascribe intentionality to procedurally generated content:

"I think it may have been procedurally generated, but I really can't say with any confidence. **There are few details to it and it doesn't have the feeling of something that I was specifically meant to find or interpret in a specific way (whereas by contrast the intentional Ozymandias reference feels hand-written and meant to be understood in a specific way because of the reference).**" (P148, Q5)

5.5 Recording Method Affordances

Different methods of recording content in games have different affordances, and scaffold the archaeological mental model. Some participants saw the recording method as a skeumorph that allowed them, or would potentially allow them, to capture the game in as much detail as possible:

"The way we are presented the game experience is without a complete overview of the village. We get one screen at a time. **I think if I were to record my experience in the game, I'd do so in map form in order to get a more complete view.**" (P174, Q6b)

"I made a map in my notebook of the space... **It definitely structured the experience. Rather than wandering it encouraged me to be systemic and observant of the space.**" (P201, Q6-2)

For other participants, their interest in a particular recording method was in its creative affordances that allowed for personal expression:

"I think writing and map making would be the most interesting ways of recording the experience, **because they require interpreting the evidence and crafting a story on your own, which to me is the real essence of the game.**" (P178, Q6b)

"I think the experience was quite evocative in a way that would be hard to convey with screenshots or even video. **I think creative writing would capture the experience best**" (P168, Q6b)

As seen with the response from P201 above, when we explicitly prompted players to record their game play as part of the theoretical sampling, there was some reflection that this changed their experience of the game. This reflects work done by Morgan et al on

archaeologists' mental models being supported by drawing methods [62]. In addition, the record could be a form of interpretation in of itself, and a way of communicating this interpretation with other players.

6 Discussion

6.1 Building on Existing Models

6.1.1 Archaeological Gameworld Mental Models and Interpretive Agency. Our theory of an archaeological gameworld mental model builds on and compliments Cole and Gillie's [18] concept of interpretive fictional agency. They explicitly discuss how players couched their interpretive experience of a game through how they interpreted environmental cues. However, Cole and Gillie's model does not engage with how and on what basis players form interpretations about games based on environmental storytelling. This is in part because they see interactive fictional agency as "an ability rather than the result of a process," and they only focus on games that have a pre-existing narrative, rather than those with emergent narratives like *Nothing Beside Remains*. The archaeological gameworld mental model is a new theory that can be employed to understand the process of interpretive agency, not just predetermined outputs. We propose the term "archaeological interpretive agency" to more specifically refer to games with emergent narratives as speculated through environmental storytelling.

6.1.2 Environmental Storytelling as an Archaeology of the Gameworld. Tying together the strands of environmental storytelling and procedurally generated gameworlds that we set out at the beginning of this paper, and in answer to RQ2, we found that participants engaged not only in an archaeology of the fictional game space, but an archaeology of the underlying generator as well. We saw this through the identification of procedural gameworld affordances — participants attempted to identify procedurally generated content based on how well they thought it could imitate hand-written content. This two-tier archaeological interpretation also correlates with Jørgensen's [46] definition of a gameworld that encapsulates not just the playable game space but the underlying system that operates it.

6.1.3 Recording Method Affordances. Our theory posits that recording method affordances scaffold the archaeological gameworld mental model by externalising information. Recording method mechanics are common across many different games, such as the range of game map interfaces identified in Dugas et al's study [84]. We can also see how Dugas et al's concept of atomic affordances for map interfaces complements this theory, as the smaller affordances of a game map interface contribute to its overall ability to scaffold the gameplay experience. However, Dugas et al do not examine how these affordances may help players form interpretations about environmental storytelling. Furthermore, the aggregation affordances that Kaptelinin and Nardi [47] discuss, describing how technological artefacts can be combined, is highly relevant to our idea of recording method affordances in which both digital and non-digital recording methods can help to scaffold the archaeological gameworld mental model. This work also complements the study of maps and other tools to support mental models for spatial reasoning in psychology [85].

6.2 Interdisciplinary Contributions

6.2.1 HCI, Archaeology and Affordances. Though this was not the focus of our study, we believe that our theory and the supporting literature demonstrates the potential for further interdisciplinary studies on affordances drawing from both HCI and archaeology. As Kaptelinin and Nardi [47] have pointed out, Gibson’s original theory of affordances is not appropriate for complex technologies. However, archaeology as a discipline has long grappled with the affordances of material culture and tools, as well as digital materiality [61], but has to date had little impact on wider scholarship regarding the material turn and the digital. Our theory of an archaeological gameworld mental model invites wider application of archaeological theory to digital material culture in HCI, especially in terms of attribution of context and assemblage affordances.

6.2.2 Non-Domain Specialists as Gameworld Archaeologists. With RQ1, we posed the question of how players form interpretations based on environmental storytelling in *Nothing Beside Remains*. Through our grounded theory analysis, we identified that participants in the study formed interpretations based on gameworld environments in ways that have parallels with archaeological interpretation and theory. For example, players would assume a particular context for an object, such as a barn, based on the building layout and the nearby presence of plants. Furthermore, participants made interpretations based on object assemblages, such as a room with several chairs being interpreted as a living area. This shows that even non-domain specialists can be understood as gameworld archaeologists. As Shanks [77] defines the “archaeological imagination” to be not just rooted in archaeology itself but also wider cultural practices of historical fiction, personal memory-making and archives, archaeological theory is a productive lens through which to understand both the craft and interpretation of environmental storytelling.

6.2.3 Archaeological Performance and Game Preservation. Player interpretations of environmental storytelling could be conceived as a form of archaeological roleplay, which just like analogue archaeology, is a site specific performance [7] informed by the affordances of a place (or in the case of a game, the gameworld affordances). Our identification of recording method affordances scaffolding a player’s archaeological gameworld model resonates with Shanks and Pearson’s [70] emphasis on the importance of documenting both archaeological fieldwork and theatrical performance through photography, plans and other methods. More broadly, we would argue that archaeological theory and method can contribute to HCI, encouraging player recordings not only as a means to study player experience but also as a form of game preservation. In tandem, HCI studies can provide insights into designing gameworlds that elicit interpretive agency and the production of interpretive documentation by players, which has broader application in public archaeology and audience engagement.

6.3 Applications for Designers and Researchers

6.3.1 Archaeological Gameworld Mental Models and Information Games. As mentioned in the related work section, *Nothing Beside Remains* can be classed as an information game. Game developer Tom Francis coined the term to refer to titles where the main goal

is to acquire information and come up with theories based on the information one has, to in turn be able to acquire more information [31]. Games in this genre include *Return of the Obra Dinn* [56], *Heaven’s Vault* [42], *Outer Wilds* [27] and *Her Story* [59]. As Cook has pointed out, one of the key design challenges for information games is modelling player knowledge [21]. Eliciting player archaeological gameworld mental models, potentially through recording methods such as photography or map-making, could be an avenue for designers of information games and puzzle games to further explore as a creative mechanic and tool for testing player assumptions.

6.3.2 Narrative Design Implications for Interpretive Difficulty. As mentioned above, this work builds on the model of interpretive agency. We also believe it to be applicable to the concept of interpretive difficulty [43] in games, which can be recognised in titles such as *Elden Ring* [33] that are renowned for their obtuse world-building as relayed to players through environmental storytelling. Caracciolo has identified themes of uncertain materiality and interpretive open-endedness in his study of archaeological fandom and *Elden Ring* [9] and this chimes with our concept of archaeological gameworld affordances that invite players to interpret their material remains. These affordances, incongruity, temporal materiality, ambiguity and intentionality, are useful for narrative designers who want to engage players with interpretive difficulty. In addition to this, we have argued that these qualities also overlap with procedural affordances in a way that is particularly powerful in terms of environmental storytelling. The uncanny syntax of PCG [50], allows for such instances of incongruous or ambiguous content. We propose the term “procedural interpretive difficulty” to refer to this ability for procedurally generated content to provide compelling prompts for archaeological interpretive agency.

6.3.3 Grounded Theory and Iterative Game Design. Grounded theory analyses tend to be conducted on small samples of participants as part of semi-structured interviews [3][19][17]. Our study prioritised breadth over depth of responses in analysing set survey responses. However, and as our results demonstrate, participants were able to engage on their own terms and produce extensive and rich written responses to the survey questions. Our study and its resulting theory suggests a novel methodology for HCI researchers wishing to study games in the future. In particular, using grounded theory on participant responses to a game that we ourselves had developed allowed us to incorporate the results of the initial survey into the iterative design of *Nothing Beside Remains*, while also being able to bring these design changes to the theoretical sampling. We believe this “design-grounded perspective” [52] has great potential for future HCI studies on games using grounded theory.

6.4 Limitations and Future Work

6.4.1 Sampling Limitations and Assemblages of Play. In this work we focus on a single game, without follow-up interviews. By recruiting participants to complete an anonymous online survey, we did not interact with them face to face and pick up on how the context in which they played the game may have also influenced their experience. This ties in with Taylor’s concept of the “assemblage of play,” [83] that the hardware and physical context in which a

player accesses a game are an integral part of play, not just what is rendered on-screen.

As we argue in §4.3, there were distinct advantages to using our own game as the case study for this work, especially in terms of being able to create bespoke builds. The remote approach also had advantages in that participants could engage with the game at their own pace, with less external pressure, which is important for such a reflective and open-ended task. However, we intend to extend this work by conducting further studies incorporating in-person interviews in which we ask participants to produce analogue records of their experience. We have already conducted one in-person pilot study where participants drew maps of the villages as they explored them, which shows promising new avenues for studying the archaeological gameworld mental model and recording method affordances.

6.4.2 Adapting Mental Models. Several HCI studies on player mental models have examined how these change over time, especially as players adapt to new information [88]. Participants in our study were only expected to play the game once, and the game state did not change over time. There was potential for their mental model to be challenged by objects they encountered in the game, as evidenced by some participants explaining how their theories changed over time.

The single-round nature of our study is in line with the original premise of the game on its initial release, which instructed players to only play it once. Several participants commented on a desire to replay the game in order to understand the generative system better, however, so a potential extension of the study could measure players' changing mental models over the course of exploring multiple villages across several sessions of *Nothing Beside Remains*.

6.4.3 Quantitative Analysis. We reported on the rich qualitative data of our study, however we also collected telemetric data of participants' movements, interactions and sight lines during play. We plan to conduct a quantitative analysis of this data that can be cross-referenced with participant interpretations, potentially giving further insight into how responses may have been shaped by the path they took through the game landscape. In particular, it might provide insight into how theories formed over time or were influenced by the order in which different areas of the map were explored.

7 Conclusion

In this paper we present the first large-scale grounded theory analysis of how players interpret environmental storytelling in a procedural game world, using *Nothing Beside Remains* as a case study. We propose the archaeological gameworld mental model as a way of understanding not only how participants interpreted the playable game space, but theorised about the underlying procedural system that generated it. This model also accounts for prior knowledge of analogue and digital culture, the constraints of the playable gameworld, archaeological gameworld affordances that prompt a player to engage in further interpretive activity and recording method affordances that shape and reaffirm the model itself. This interdisciplinary work draws on theory from archaeology, HCI and procedural design, and has implications for the broader study of

interpretive agency in games and recording player interpretations as a form of games preservation.

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A Survey Questions

The following is the full list of survey questions in the initial survey. We have omitted questions regarding consent for data collection.

- Please enter the 6-character code that was on the screen when you opened the game (e.g. PVY78B in the example below) so that we can connect your responses to your play data.
- Is this the first time you have played *Nothing Beside Remains*?
 - Yes
 - No
- Please check all of the following that apply to you:
 - I play videogames in my spare time
 - I work in the games industry
 - I am a researcher who uses games as part of my work
- What is your age?
- How would you define “procedural content generation”? If you are not familiar with the term you can leave this field blank.
- Write your interpretation of what happened to the village in the game. What reasons do you have for this interpretation?
- Do you have any alternative interpretations? If so, explain your reasoning.
- Could you distinguish between procedurally generated and hand-written content, and if so how?
- Pick one object from the game and describe how you think it came to be left where you found it in the village. Why did you choose this object?
- Do you think the object you chose for question 4 was procedurally generated? Why or why not?
- If you were to create a record of your experience in the game, which of the following methods would most appeal to you:
 - Creative writing
 - Screenshots
 - Video footage
 - Map-making
 - Other (specify)

- Why did you choose the above method(s)? If you were to share the record with others would that change your answer?
- If you made a record of your experience playing the game, what method did you use and why? Did it affect your experience or interpretation of the game?

For the second survey, modifications were made to the question about the participant’s background:

- Please check all of the following that apply to you:
 - I play videogames in my spare time
 - I work in the games industry
 - I am a researcher who uses games as part of my work
 - I often record my gameplay experiences in some form
 - I have a personal interest in archaeology and/or heritage
 - I am a researcher who studies the past as part of my work

Additionally, the final two questions were replaced with a single new question:

- If you made a record of your experience playing the game, what method did you use and why? Did it affect your experience or interpretation of the game?

B Coding

The table on the following page shows open codes and the corresponding focused codes under which they were grouped. Codes from theoretical sampling are **in bold text** and are followed by an asterisk (*).

Focused Codes	Open Codes
Skeuomorph of human labour	Designed; Focus on designer input; Reducing workload; Labour conditions; Hand-crafted vs algorithmic; Designer oversight; Imitating human labour; Finite input vs infinite output; Algorithmic vs hand-designed; Indirect authorship; Extending human labour; Lack of human input; Evidence of human design; Symmetry; Perceptual uniqueness; Lack of human input* ; Focus on output* ; Deism* ; Quality of human labour* ; Accurate skeuomorph* ; Perceptibly unique* ; Perceptibly unique content* ; Perceptibly unique output* ; Perceptual uniqueness* ; Perceptibly unique object* ;
Imperfect skeuomorph	Coherency; Observable patterns; Curation; Aesthetic of randomness; Procedural aesthetic; Incongruence; Designed repetition; Hand-crafted object vs procedural placement; Perceptibly modular content; Low quality; Procedural logic; Procedural syntax; Identification of glitch; Lack of cohesion; Identical objects; Mundane vs bespoke content; All content generated; Fictional significance; Lack of external reference; Thematic relevance; Interpretation affected by appearance of tiles; Lack of cohesion; Identical objects; Handwritten assumed to be procedural; Writing as enhancing PCG; Procedural content requiring interpretive filter; Lack of intentionality* ; Narrativisation of procedural syntax* ; Diagnostic bugs* ; Perceptibly modular content* ; Usage errors* ; Lack of cohesion* ; Procedural uncanny* ; Perceived repetition* ; Human skeuomorph of PCG* ; Perceptibly similar content* ; Unique content not procedural* ; Procedural uncanny* ; Lack of consistency* ;
Procedural affordances	Randomised; Rule-based; Examples of games assets; Enhancing player experience; Novel content; Cyclical generation; Focus on novelty experience; Generation at time of gameplay; Spatial content; Focus on player experience; Spatial generation; Game example; Impermanent; Repetitive novelty; Focus on output; Use of constraints; Fallacy of definition; Predictability vs unpredictability; Randomness vs constraints; Algorithmic control; Player input; Procedural tone; Affordances of PCG; Unique content vs unique experience; Map-making and procedural affordances; Rule-based* ; PCG as inherently emergent* ; Filter of the algorithm* ; System-based* ; Seed-based* ; Indeterminacy* ; Constraint-based* ; Repetition* ;
Pre-existing knowledge	Game example; Knowledge of procedural design; Spatial design conventions; Challenges of textual generation; Vibes-based; Pattern recognition through repeated generations; PCG as non-narrative; Speculation informed by life experience; Genre conventions; Reference to archaeology; External reference; Architectural conventions; Theory about generator; Pattern recognition; Interpretation affected by appearance of tiles; Fictional significance; Extrapolation from experiencing a glitch; Technical archaeological term; Caught in the glitch; Occam's razor as justification for theory; Meta knowledge of games; Need to explore entirety of map; Meta understanding as game; Reference to other game; Existing knowledge of recording method; Map-making as reference to older games; Metatextual narrative; Emulating archaeological report; Language borrowed from fanfiction; Existing archaeological experience; Lack of experience with recording method; Ease of recording method; Finds list as informed by archaeological background; Standardised itemisation; Reference to other games* ; Replayability* ; Need to replay* ; Lack of familiarity* ; Need for multiple readings* ; Pattern recognition* ; Reference to Ozymandias* ; Understanding of procedural text* ; Lack of data* ; Uncertain* ; Reference to Ozymandias* ; Habitual use of recording method* ; Pattern recognition* ; Analogy to real-world archaeology* ; Nostalgic affordances* ;
Framing	Interpretation affected by framing; Self-referential; Lack of external reference; Lack of framing device; Reference to simulation; Interpretation framed by study; Comparison between playthroughs; Questioning assumptions in text; Role-playing as an archaeologist; Game aesthetics; Need for recording to support theory; Primacy bias; Unreliable narrator; Primacy bias* ; Statue as primer* ; Framing encouraging creative speculation* ; Structured the experience* ;
Assemblage affordances	Context-based generation; Proximity as association; Assemblage of skeletons assumed to be graveyard; Multiple interpretations of altars with bones; Agricultural material culture; Narrativisation of skeletal assemblage; Importance of assemblage; Terminus ante quem; Clear association; Agri-cultural assemblages* ; Apocalyptic assemblage* ; Attribution of skeletal assemblage* ;

Intentionality	Deliberate tone; Conscious design; Random vs purposeful; Authorial intent; Glitch as hand-written; Statue as landmark; Church as identifiable landmark; Repetition as hand-written; Spatial generation vs hand-written text; Repetitive content as hand-written; Thematic relevance; Mundane vs bespoke content; Randomness vs intentionality; Attribution of intentionality; Lack of proper burial; Evidence of ongoing memorialisation; Broken material culture; Evidence of maintenance; Voluntary or involuntary; Intentional breakage; Intentional toppling of statue; Evidence of design; Intentionality* ; Uniqueness as diagnostic of hand-written* ; Attribution of context as hand-written* ; Intentionality vs randomness* ; Lack of proper burial* ; Repetition as significant* ;
Environmental affordances	Reference to colour palette; Reclamation by desert; Environmental context of object; Environmental conditions; Identification of the sandstorm; Reference to sandstorm; Reason for sustained abandonment; Concept of seasonal occupation; Tool as emblematic of environmental change; Environmental conditions* ; Sound effect attributed to environment* ;
Affordances of fictional objects	Survival bias; Reference to imagery on the pews; Speculation about repurposed materials; Assumed portability of objects; Affordances of materials; Metal objects as evidence of craft specialisation or trade; Metal objects as evidence of invading culture; Preservation of scents; Evidence of craft specialisation; Longevity of statue; Assumed portability of objects* ; Technological affordances of materials* ; Affordances of objects* ; Reference to ASCII symbol* ;
Attribution of context	Context-based generation; Use of cardinal directions; Identification of cemetery with the church; Identification of zoning; Attribution of mortuary contexts; Location of skeletons; Reference to layout; Attribution of context; Building context; Generation context-dependent* ; Context-based generation* ; Attribution of grave site* ; Attribution of context* ;
Ambiguity	Lack of textual evidence; Reference to number 7; Lack of data; Repetition of the number 7; Need for more data; Need to collect more data; Uncanny; Fallibility of memory; Emergent narrative from limited evidence; Ambiguity of recording method inspiring creativity ; Ambiguity of game inviting creative writing; Repetition of number 7* ;
Temporal materiality	Lack of evidence for wear over time; Evidence of recent abandonment based on material remains; Lack of evidence for re-purposing; Multi-phase occupation; Condition of objects; Time since abandonment; Location and condition of skeletons; Rapid collapse; Condition of buildings; Evidence of abandonment; Nature of ruination; Evidence of slow decline; Change in original function; Location and condition of skeletons* ; Condition of objects* ; Ruination* ; Multi-phase activity* ;
Mental model of catastrophe	Inference through what is abandoned; Ending as part of a natural cycle; Natural disaster; Damage as evidence of violence; Lack of material evidence for violence; Reasoning for why material culture was abandoned; Material evidence of animal attack; Evidence of animal attack; Material evidence of violence; Natural disaster inferred from lack of proper burial; Interpretation of animals abandoned; Unusable environment interpreted as self-isolation; Lack of redevelopment; Post-depositional processes; Association between high walls and animal threat; Rapid abandonment; Evidence of surplus resources; Valuable objects left behind as indicative of catastrophe; Destruction deposit; Assumed lack of value as left behind; Expression of interior mental model; Mental model of catastrophe* ; Theory of abandonment* ; Enhanced mental model* ;
Diagnostic features	Pews as diagnostic of religious structure; Altar as evidence of worship; Reference to Christianity; Classification of building type; Ritual activity based on skeletal and material remains; Identification as religious centre; Identification of religious belief; Pews as diagnostic* ; Pew as diagnostic of religious structure* ; Identification* ;
Object as attributed to individual or culture	Identification of prestige vs non-prestige objects; Population based on material culture; Trident as icon; Pocket watch as diagnostic; Identification of name with pocket watch; Tulip divinity; Evidence of standardisation; Reference to imagery on the pews ; Interpretation of the statue inscription; Identification of sacred vs profane; In-game iconography as idealised past; In-game iconography as representational; Object as emblematic of culture; Name as denoting personal item; Association with an individual; Identification with individual; More detail as more significance; Interest in human activity over evidence of human activity; Later intrusion by other explorers; Identification of everyday object; Perceptual consistency; Perceptibly unique object as emblematic of individual* ;

Incongruity	Comparison between objects; Unique content; Perceptibly unique objects; Statue as incongruous with other material culture; Incongruity of pews with other material culture; Comparison between number of skeletons and buildings; Discrepancy between player size and buildings; Comparison between skeletons; Pocket watch as incongruous with other material culture; Perceptual uniqueness; Recording method to capture perceptibly unique content; Comparison between objects* ; Incongruity* ; incongruity of pocket watch* ; Watch as anachronistic* ; Unique pocketwatch compared to identical examples* ; Watches as incongruous but not unique* ; Unique objects more significant* ;
Analogy (creative speculation)	Analogy with real-world archaeological site; Creative speculation; Reference to fictional media; Large skeletons as fantastical; Theory of religious conflict; Theory of religion falling out favour; Poetic end; Ozymandias as reference point; Association with trident and Poseidon; Divine retribution; Narrativisation of statue; Idea of divine retribution; Theory of political stratification; Supernatural theory; Hubris; Extreme theory as comedy; Opposite interpretation as alternative; Secondary theory as more specific but less evidence; In-game iconography as idealised; Genius loci; Narrativisation of statue* ; Unnatural disaster* ; Narrativisation of procedural syntax* ; Narrativisation of statue as grave* ; Statue destruction as symbolic* ; Enjoyment of creative speculation* ; Entire village as tribute* ;
Record as skeuomorph	Recording method as skeuomorph; Capture all details without bias; Desire for in the moment narration; Fidelity of recording method; Limits of direct replication; Video and screenshot as more direct interpretation; Perceived objectivity of recording method; Aide memoire; Desire not to miss content; Affordances of recording methods; Affordances of the digital; Desire to record everything* ; Vibes vs fidelity of record* ; Reliance on memory* ; Fidelity of record* ;
Record as personal expression	Capturing feelings; Meaning from sharing; Capturing experience ; Personal vs public recording; Enjoy recording method; Difficulties of recording while playing; Recording method as breaking immersion; Recording method as personal reflection; Personal sense of achievement and progress; Recording as roleplay; Choosing method they like to consume; Difficulty recording while playing* ; Not motivated to record* ; Recording as roleplay* ; Note-taking as internal monologue* ;
Spatial affordances	Need to see entirety of map; Spatial affordances; Recording own movements; Affordances of partial view; Active use of map-making to understand the world; Map-making to understand spatial affordances; Lack of spatial affordances; Lack of narrative affordances; Spatial generation* ; Spatial affordances* ; Perceived scale of game* ;
Abstraction	Abstract visuals affording abstraction; Parsibility of abstract environment; Recording method informed by affordances of game; Annotational affordances of map; Recording method informed by mechanical affordances of the game;
Record as interpretation	Emergent narrative from theory; Emergent narrative from meaning making; Interpretation and narrativisation; Record allowing for pattern recognition; Stages of map-making; Recording methods complimenting each other; Importance of sharing process behind interpretation; Map as supporting theories; Record as enhancing interpretation; Map as interpretation; Map able to show temporal change; Desire to limit influence of own interpretation; Recorded perceived notable items* ; Recording interesting content* ;
Record as communication	Parsibility of recording method; Parsibility of record; Recording method as encouraging others to experience the game; Immersive quality of recording method; Enjoyment of audience; Visual appeal of record; Recording as aiding engagement; Dependent on purpose; Comparison between interpretations;